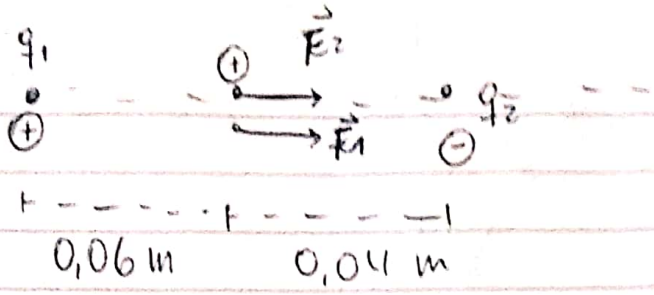
Punto a)

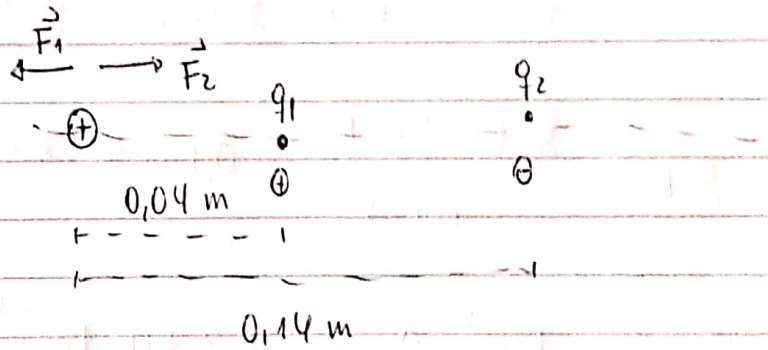


$$\vec{F_1} = 9\cdot 10^9 \frac{(5\cdot 10^{-9})(3\cdot 10^{-9})}{(0{,}06)^3} 0{,}06\,\hat{\imath} = 3{,}75\cdot 10^{-5}\,\hat{\imath}\,[N]$$

$$\vec{F_2} = 9\cdot 10^9 \frac{(5\cdot 10^{-9})(3\cdot 10^{-9})}{(0{,}04)^3} (0{,}04\,\hat{\imath}) = 8{,}44\cdot 10^{-5}\,\hat{\imath}\,[N]$$

$$\vec{F_T} = \vec{F_1} + \vec{F_2} = 1{,}219\cdot 10^{-4}\,\hat{\imath}\,[N].$$

Punto b)



$$\vec{F_1} = 9\cdot 10^9 \frac{(5\cdot 10^{-9})(3\cdot 10^{-9})}{(0{,}04)^3} 0{,}04\,(-\hat{\imath})$$

$$= 8{,}438\cdot 10^{-5}\,(-\hat{\imath})\,[N]$$

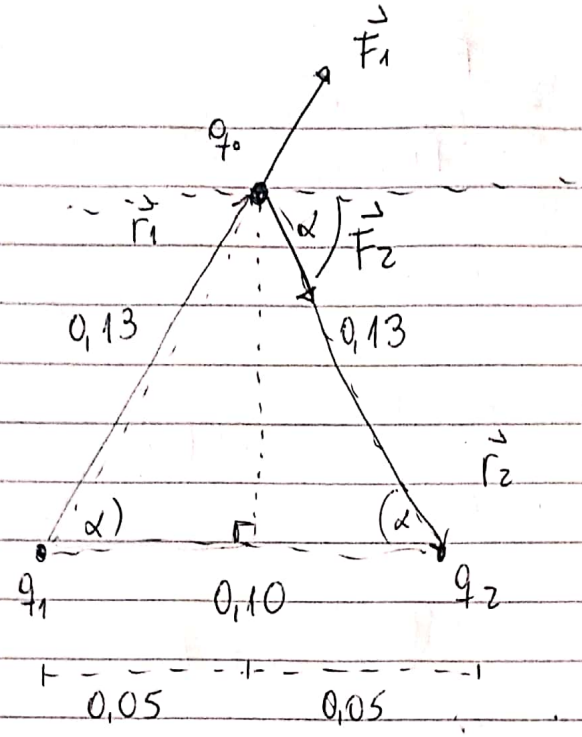
$$\vec{F_2} = 9\cdot 10^9 \frac{(5\cdot 10^{-9})(3\cdot 10^{-9})}{(0{,}14)^3} 0{,}14\,\hat{\imath}$$

$$= 6{,}888\cdot 10^{-6}\,(\hat{\imath})\,[N]$$

$$\vec{F_T} = \vec{F_1} + \vec{F_2} = 7{,}749\cdot 10^{-5}\,(-\hat{\imath})\,[N]$$

neado con CamScanner

Punto c)



$$\cos\alpha = \frac{0{,}05}{0{,}13}$$

$$\alpha = 67{,}38°$$

$$\vec{r}_1 = 0{,}13\cos 67{,}38\,\hat{\imath} + 0{,}13\,\text{sen}\,67{,}38\,\hat{\jmath}$$

$$\vec{r}_2 = 0{,}13\cos 67{,}38\,\hat{\imath} - 0{,}13\,\text{sen}\,67{,}38\,\hat{\jmath}$$

$$\Rightarrow \quad \vec{F}_1 = 9\cdot 10^9 \frac{(5\cdot 10^{-9})(3\cdot 10^{-9})}{(0{,}13)^3}(0{,}13)[\cos\alpha\,\hat{\imath} + \text{sen}\,\alpha\,\hat{\jmath}]$$

$$= 3{,}072\cdot 10^{-6}\,\hat{\imath} + 7{,}374\cdot 10^{-6}\,\hat{\jmath}\ [N].$$

$$\vec{F}_2 = 9\cdot 10^9 \frac{(5\cdot 10^{-9})(3\cdot 10^{-9})}{(0{,}13)^3}(0{,}13)[\cos\alpha\,\hat{\imath} - \text{sen}\,\alpha\,\hat{\jmath}]$$

$$= 3{,}072\cdot 10^{-6}\,\hat{\imath} - 7{,}374\cdot 10^{-6}\,\hat{\jmath}\ [N]$$

$$\vec{F}_T = \vec{F}_1 + \vec{F}_2$$

$$= 6{,}144\cdot 10^{-6}\,\hat{\imath}\ [N]$$

Escaneado con CamScanner